

STRINGS

TRIM

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the lab name, and the user's name and ID. The left sidebar shows the 'TRIM' problem description: 'Write a Java program to trim a given string and remove leading and trailing spaces.' The main area is the 'PROGRAM EDITOR' with a Java code editor containing the following code:

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         String str = sc.nextLine();
7         String trimmedStr = str.trim();
8         System.out.println(trimmedStr);
9     }
10 }
```

Below the editor is the 'OUTPUT' section, which displays 'java programming'. The right sidebar shows the 'INPUT' section with 'java programming' entered.

SPLIT

The screenshot shows the SIMATS Online Programming Lab interface for the 'SPLIT' problem. The left sidebar shows the problem description: 'Write a Java program to split a given string into an array of substrings based on a specified delimiter.' The main area is the 'PROGRAM EDITOR' with a Java code editor containing the following code:

```
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.println("Enter the string:");
8         String str = sc.nextLine();
9
10        System.out.println("Enter the delimiter:");
11        String delimiter = sc.nextLine();
12
13        String[] parts = str.split(delimiter);
14
15        System.out.println("Substrings after split:");
16        for (String part : parts) {
17            System.out.println(part);
18        }
19
20        sc.close();
21    }
22 }
```

Below the editor is the 'OUTPUT' section, which displays the following text:

```
Enter the string:
Enter the delimiter:
Substrings after split:
java
```

The right sidebar shows the 'INPUT' section with 'javaprogramming programming' entered.

CONVERT

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QUESTIONS 20 / 20 completed

Convert

CONVERT
Write a Java program to convert a given string to a char array.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         System.out.print("Enter a string: ");
7         String str = sc.nextLine();
8
9         char[] charArray = str.toCharArray();
10
11         System.out.print("Char array: ");
12         for (char c : charArray) {
13             System.out.print(c + " ");
14         }
15     }
16 }
```

INPUT
java

OUTPUT
Enter a string: Char array: j a v a

UPPER

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QUESTIONS 20 / 20 completed

Upper

UPPER
Write a Java program to convert a given string to uppercase.

PROGRAM EDITOR Java Run Save

```
1 public class upper
2 {
3     public static void main(String[] args)
4     {
5         String s="java";
6         System.out.println("upper case :"+ s.toUpperCase());
7     }
8 }
```

INPUT
Your INPUT goes here!

OUTPUT
upper case :JAVA

RM-CHAR

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and the user's name and ID. The left sidebar shows the 'QUESTIONS' section with '20 / 20 completed' and a dropdown menu set to 'RM-Char'. Below this, the 'RM-CHAR' problem description is shown: 'Write a Java program to remove a specified character from a given string.' The main area is the 'PROGRAM EDITOR' with a 'Java' language selector and 'Run' and 'Save' buttons. The code editor contains the following Java code:

```
1 public class main{
2     public static void main(String[] args){
3         String s="hello";
4         char ch='l';
5         String result=s.replace(String.valueOf(ch),"");
6         System.out.println(result);
7     }
8 }
```

Below the code editor is the 'OUTPUT' section, which currently shows 'he o'. To the right of the code editor is the 'INPUT' section, which contains the text 'Your INPUT goes here!'.

REPLACE

The screenshot shows the SIMATS Online Programming Lab interface for the 'REPLACE' problem. The header is the same as the previous screenshot. The left sidebar shows the 'QUESTIONS' section with '20 / 20 completed' and a dropdown menu set to 'Replace'. Below this, the 'REPLACE' problem description is shown: 'Write a Java program to replace a specified character with another character in a given string.' The main area is the 'PROGRAM EDITOR' with a 'Java' language selector and 'Run' and 'Save' buttons. The code editor contains the following Java code:

```
1 import java.util.Scanner;
2
3 public class main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         String str = sc.nextLine();
7         char oldChar = sc.nextLine().charAt(0);
8         char newChar = sc.nextLine().charAt(0);
9
10        String result = str.replace(oldChar, newChar);
11
12        System.out.println(result);
13    }
14 }
```

Below the code editor is the 'OUTPUT' section, which currently shows 'hollo'. To the right of the code editor is the 'INPUT' section, which contains the text 'hello', 'e', and 'o' on separate lines.

COMPARE



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QUESTIONS 20 / 20 completed

Compare

COMPARE

Write a Java program to compare two strings lexicographically.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter first string: ");
8         String s1 = sc.nextLine();
9
10        System.out.print("Enter second string: ");
11        String s2 = sc.nextLine();
12
13        int result = s1.compareTo(s2);
14
15        if (result < 0) {
16            System.out.println("First string is lexicographically smaller than second
17        } else if (result > 0) {
18            System.out.println("First string is lexicographically greater than second
19        } else {
20            System.out.println("Both strings are lexicographically equal.");
21        }
22    }
23 }
```

OUTPUT

Enter first string: Enter second string: First string is lexicographically smaller than second string.

INPUT

hello
java

FIND



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QUESTIONS 20 / 20 completed

Find

FIND

Write a Java program to find the index of a given character or substring in a given string.

PROGRAM EDITOR

Java Run Save

```
1 public class main
2 {
3     public static void main(String[] args)
4     {
5         String str ="java is aprogramming language";
6         String a="language";
7         int index=str.indexOf(a);
8         if(index>=0)
9         {
10            System.out.println("found index at : " + index);
11        }else{
12            System.out.println("element is not found");
13        }
14    }
15 }
```

OUTPUT

found index at : 21

INPUT

Your INPUT goes here!

SUFFIX

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QUESTIONS 20 / 20 completed

Suffix

SUFFIX
Write a Java program to check if a given string ends with a specific suffix.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Input: ");
8         String str = sc.nextLine();
9
10        System.out.print("Suffix: ");
11        String suffix = sc.nextLine();
12
13        if (str.endsWith(suffix)) {
14            System.out.println("True");
15        } else {
16            System.out.println("False");
17        }
18
19        sc.close();
20    }
21 }
```

OUTPUT

Input: Suffix: False

INPUT
helloworld
hello

PALINDROME

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QUESTIONS 20 / 20 completed

Palindrome

PALINDROME
Write a Java program to check if a given string is a palindrome or not.

PROGRAM EDITOR Java Run Save

```
1 public class main
2 {
3     public static void main(String[] args)
4     {
5         String s="java";
6         String rev=new StringBuilder(s).reverse().toString();
7         if(s.equals(rev))
8             System.out.println("palindrome");
9         else
10            System.out.println("not a palindrome");
11     }
12 }
```

OUTPUT

not a palindrome

INPUT
Your INPUT goes here!

OCCURRENCE

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QUESTIONS 20 / 20 completed

Occurrence

OCCURRENCE
Write a Java program to count the number of occurrences of a given character in a string.**PROGRAM EDITOR** Java Run Save

OUTPUT

INPUT

CHECK

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QUESTIONS 20 / 20 completed

Check

CHECK
Write a Java program to check if a given string contains only digits.**PROGRAM EDITOR** Java Run Save

OUTPUT

INPUT

LENGTH

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and the user's name and ID. The left sidebar shows the 'LENGTH' problem description: 'Write a Java program to find the length of the longest substring without repeating characters in a given string.' The main area is the 'PROGRAM EDITOR' with a Java code editor containing a solution for finding the length of the longest substring without repeating characters. The 'INPUT' field on the right contains 'JOY IS SMART'. The 'OUTPUT' field at the bottom shows the result '6'.

QUESTIONS 20 / 20 completed

Length

LENGTH

Write a Java program to find the length of the longest substring without repeating characters in a given string.

PROGRAM EDITOR

```
1 import java.util.HashMap;
2 import java.util.Scanner;
3
4 public class Main {
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7         String s = sc.nextLine();
8         System.out.println(lengthOfLongestSubstring(s));
9     }
10
11     static int lengthOfLongestSubstring(String s) {
12         HashMap<Character, Integer> map = new HashMap<>();
13         int maxLength = 0;
14         int start = 0;
15
16         for (int end = 0; end < s.length(); end++) {
17             char currentChar = s.charAt(end);
18             if (map.containsKey(currentChar) && map.get(currentChar) >= start) {
19                 start = map.get(currentChar) + 1;
20             }
21             map.put(currentChar, end);
22         }
23         return maxLength;
24     }
25 }
```

INPUT

JOY IS SMART

OUTPUT

6

LOWER

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and the user's name and ID. The left sidebar shows the 'LOWER' problem description: 'Write a Java program to convert a given string to lowercase.' The main area is the 'PROGRAM EDITOR' with a Java code editor containing a solution for converting a string to lowercase. The 'INPUT' field on the right contains 'Your INPUT goes here!'. The 'OUTPUT' field at the bottom shows the result 'Lower case : java'.

QUESTIONS 20 / 20 completed

Lower

LOWER

Write a Java program to convert a given string to lowercase.

PROGRAM EDITOR

```
1 public class main
2 {
3     public static void main(String[] args)
4     {
5         String s="JAVA";
6         System.out.println("Lower case : " + s.toLowerCase());
7     }
8 }
```

INPUT

Your INPUT goes here!

OUTPUT

Lower case : java

REMOVE

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and the user's name and ID. The left sidebar shows the 'REMOVE' problem description: 'Write a Java program to remove all white spaces from a given string.' The main area contains a 'PROGRAM EDITOR' with a Java code snippet that reads input, removes all white spaces using a regular expression, and prints the result. The 'INPUT' field contains 'JUNGLE IS DARK' and the 'OUTPUT' field shows 'JUNGLEISDARK'.

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QUESTIONS 20 / 20 completed

Remove

REMOVE

Write a Java program to remove all white spaces from a given string.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         String str = sc.nextLine();
7
8         String result = str.replaceAll("\\s", "");
9
10        System.out.println(result);
11    }
12 }
```

INPUT

JUNGLE IS DARK

OUTPUT

JUNGLEISDARK

REVERSE

The screenshot shows the SIMATS Online Programming Lab interface for the 'REVERSE' problem. The left sidebar shows the 'REVERSE' problem description: 'Write a Java program to reverse a given string.' The main area contains a 'PROGRAM EDITOR' with a Java code snippet that reverses the input string using a StringBuilder. The 'INPUT' field contains 'Your INPUT goes here!' and the 'OUTPUT' field shows 'aVaj'.

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QUESTIONS 20 / 20 completed

Reverse

REVERSE

Write a Java program to reverse a given string.

PROGRAM EDITOR Java Run Save

```
1 public class main
2 {
3     public static void main(String[] args)
4     {
5         String s="java";
6         String rev=new StringBuilder(s).reverse().toString();
7         System.out.println(rev);
8     }
9 }
```

INPUT

Your INPUT goes here!

OUTPUT

aVaj

CONCATINATE

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and the user's name and ID. The left sidebar shows the 'CONCATINATE' problem selected. The main area contains a 'PROGRAM EDITOR' with a Java code snippet that concatenates 'Java' and 'Programming' into 'Java Programming'. The 'OUTPUT' section shows the result of running the code.

QUESTIONS 20 / 20 completed

Concatenate

CONCATINATE

Write a Java program to concatenate two given strings.

PROGRAM EDITOR Java Run Save

```
1 public class main
2 {
3     public static void main(String[] args)
4     {
5         String s1="Java";
6         String s2="Programming";
7         String s3=s1.concat(" ".concat(s2));
8         System.out.println(s3);
9     }
10 }
```

OUTPUT

Java Programming

INPUT

Your INPUT goes here!

REPEAT

The screenshot shows the SIMATS Online Programming Lab interface for the 'REPEAT' problem. The left sidebar shows the 'REPEAT' problem selected. The main area contains a 'PROGRAM EDITOR' with a Java code snippet that finds the first non-repeated character in the string 'JUNGLE IN THE RAIN'. The 'OUTPUT' section shows the result of running the code.

QUESTIONS 20 / 20 completed

Repeat

REPEAT

Write a Java program to find the first non-repeated character in a given string.

PROGRAM EDITOR Java Run Save

```
2 import java.util.Scanner;
3
4 public class Main {
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7         String str = sc.nextLine();
8         HashMap<Character, Integer> charCount = new HashMap<>();
9         for (int i = 0; i < str.length(); i++) {
10             char c = str.charAt(i);
11             charCount.put(c, charCount.getOrDefault(c, 0) + 1);
12         }
13         char result = '\0';
14         boolean found = false;
15         for (int i = 0; i < str.length(); i++) {
16             char c = str.charAt(i);
17             if (charCount.get(c) == 1) {
18                 result = c;
19                 found = true;
20                 break;
21             }
22         }
23     }
```

OUTPUT

First non-repeated character: J

INPUT

JUNGLE IN THE RAIN

FREQUENT

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QUESTIONS 20 / 20 completed

Frequent

Write a Java program to find the most frequent word in a given string.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String input = sc.nextLine();
6         String[] words = input.trim().toLowerCase().split("\\s+");
7
8         Map<String, Integer> freqMap = new HashMap<>();
9         for (String w : words) {
10             if (!w.isEmpty()) {
11                 freqMap.put(w, freqMap.getOrDefault(w, 0) + 1);
12             }
13         }
14
15         String mostFrequent = "";
16         int maxCount = -1;
17
18         for (Map.Entry<String, Integer> entry : freqMap.entrySet()) {
19             if (entry.getValue() > maxCount) {
20                 maxCount = entry.getValue();
21                 mostFrequent = entry.getKey();
22             }
23         }
24     }
25 }
```

OUTPUT

Most frequent word: java
Frequency: 1

INPUT

java is programming
language in java

SUBSTRING

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QUESTIONS 20 / 20 completed

Substring

Write a Java program to create a substring from a given string.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Scanner;
2
3 public class SubstringExample {
4     public static void main(String[] args) {
5         Scanner scanner = new Scanner(System.in);
6         System.out.println("Enter the string:");
7         String inputString = scanner.nextLine();
8         System.out.println("Enter the starting index:");
9         int startIndex = scanner.nextInt();
10        System.out.println("Enter the ending index:");
11        int endIndex = scanner.nextInt();
12
13        try {
14            String sub = inputString.substring(startIndex, endIndex);
15            System.out.println("The substring is: " + sub);
16        } catch (IndexOutOfBoundsException e) {
17            System.out.println("Error: Invalid indices provided!");
18        }
19
20        scanner.close();
21    }
22 }
```

OUTPUT

Enter the string:
Enter the starting index:
Enter the ending index:
The substring is: ello

INPUT

hello world
1
5